

Safety data sheet

Page: 1/12

BASF Safety data sheet

Date / Revised: 02.12.2025

Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

1. Substance/preparation and manufacturer/supplier identification

Product name:

Sodium Sulfite anhydrous food grade (E221)

Use: food additive(s)

Recommended use: inorganic reducing agents, initial product for chemical syntheses, process chemical

Manufacturer/supplier:

BASF Australia Limited (ABN 62 008 437 867)

Level 23, 40 City Road, Southbank

Victoria 3006, AUSTRALIA

Telephone: +61 3 8855-6600

Emergency information:

BASF Emergency Advice Number: 1800 803 440 (24h) [within Australia]

BASF Emergency Advice Number: + 61 3 8855 6666 [outside Australia]

2. Hazard identification

Classification of the substance and mixture:

Acute toxicity: Cat.5 (oral)

Hazardous to the aquatic environment - acute: Cat.3

Label elements and precautionary statement:

Signal Word:

Warning

Hazard Statement:

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

H303 May be harmful if swallowed.
H402 Harmful to aquatic life.

Precautionary Statements (Prevention):

P273 Avoid release to the environment.

Precautionary Statements (Response):

P312 Call a POISON CENTER or physician if you feel unwell.

Precautionary Statements (Disposal):

P501 Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:

No specific dangers known, if the regulations/notes for storage and handling are considered. If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

Contact with acids liberates toxic gas.

3. Composition/information on ingredients

Chemical nature

Substance nature: Substance

sodium sulphite

CAS Number: 7757-83-7

Na₂SO₃

E 221

Hazardous ingredients

sodium sulphite

Content (W/W): >= 75 % - <= 100 %

Acute Tox.: Cat. 5 (oral)

%

Aquatic Acute: Cat. 3

CAS Number: 7757-83-7

4. First-Aid Measures

General advice:

Remove contaminated clothing.

If inhaled:

If difficulties occur after dust has been inhaled, remove to fresh air and seek medical attention. After inhalation of decomposition products: Immediately administer a corticosteroid from a controlled/metered dose inhaler. Seek medical attention.

On skin contact:

Wash thoroughly with soap and water

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

On contact with eyes:

Immediately wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., Many individuals are sensitive to sulphite additives and may experience a range of symptoms, including dermatitis, urticaria, angio-oedema, abdominal pain, diarrhoea, bronchoconstriction and anaphylaxis.

Hazards: Risk of sulfur dioxide formation by reaction with gastric acid after swallowing.

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:

water spray, carbon dioxide, foam, dry powder

Unsuitable extinguishing media for safety reasons:

water jet

Additional information:

Product will not burn.

Use extinguishing measures to suit surroundings.

Specific hazards:

Sulphur dioxide

The substances/groups of substances mentioned can be released if the product is involved in a fire.

Special protective equipment:

Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:

Product itself is non-combustible; fire extinguishing method of surrounding areas must be considered. Contaminated extinguishing water must be disposed of in accordance with official regulations. In case of fire and/or explosion do not breathe fumes.

6. Accidental Release Measures

Personal precautions:

Avoid contact with the skin, eyes and clothing. Use personal protective clothing. Ensure adequate ventilation. Avoid dust formation.

Environmental precautions:

Do not discharge into drains/surface waters/groundwater. Do not discharge into the subsoil/soil. Retain and dispose of contaminated wash water.

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Methods for cleaning up or taking up:

Sweep/shovel up. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Handling

Use only in well-ventilated areas. Avoid dust formation. Avoid contact with skin and eyes.

Protection against fire and explosion:

The substance/product is non-combustible. No special precautions necessary.

Storage

Segregate from acids and acid forming substances. Segregate from oxidants.

Suitable materials for containers: Stainless steel 1.4541, Stainless steel 1.4571, High density polyethylene (HDPE), Low density polyethylene (LDPE), Carbon steel (Iron)

Further information on storage conditions: Keep away from heat. Keep container tightly closed in a cool, well-ventilated place. Keep container dry. The product consumes oxygen. Danger of lack of oxygen in containers and tanks.

8. Exposure controls and personal protection

Components with occupational exposure limits

No occupational exposure limits known.

The nuisance dust limit value is to be kept.

The substance mentioned develops if the regulation/notes for storage and handling are not observed.

Sulphur dioxide, 7446-09-5;

STEL value 0.25 ppm (ACGIHTLV)

TWA value 5.2 mg/m³ ; 2 ppm (AU NOEL)

STEL value 13 mg/m³ ; 5 ppm (AU NOEL)

Personal protective equipment

Respiratory protection:

Breathing protection if dusts are formed. Suitable respiratory protection for lower concentrations or short-term effect: Particle filter with low efficiency for solid particles (e.g. EN 143 or 149, Type P1 or FFP1) Breathing protection if gases/vapours are formed. Combination filter for gases/vapours of organic, inorganic, acid inorganic, alkaline compounds and toxic particles (e. g. EN 14387 Type ABEK-P3)

Hand protection:

Chemical resistant protective gloves

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Suitable materials also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1):

e.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), polyvinylchloride (0.7 mm) and other

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:

Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:

Handle in accordance with good industrial hygiene and safety practice. Do not inhale vapours or dust. Hands and/or face should be washed before breaks and at the end of the shift.

9. Physical and Chemical Properties

Form:	powder, crystalline	
Colour:	white to slightly yellow	
Odour:	odourless	
Odour threshold:	Not determined due to potential health hazard by inhalation.	
pH value:	8.5 - 10.5 (5 %(m), 20 °C)	(OECD Guideline 122)
melting point (decomposition):	The substance / product decomposes.	
Boiling point:	(1,013.25 hPa) Study scientifically not justified.	
Flash point:	Study scientifically not justified.	
Evaporation rate:	The product is a non-volatile solid.	
Flammability (solid/gas):	Study scientifically not justified.	(other)
Lower explosion limit:	For solids not relevant for classification and labelling.	
Upper explosion limit:	For solids not relevant for classification and labelling.	
Thermal decomposition:	500 °C	

BASF Safety data sheet
 Date / Revised: 02.12.2025
 Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Self heating ability:	It is not a substance capable of spontaneous heating.	
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	
Fire promoting properties:	Based on its structural properties the product is not classified as oxidizing.	
Vapour pressure:	Study scientifically not justified.	
Density:	2.633 g/cm3 (20 °C) Literature data.	
Relative density:	2.63 (20 °C) Literature data.	
Bulk density:	1,400 - 1,600 kg/m3	(other)
Relative vapour density (air):	not determined	
Solubility in water:	Literature data. 220 g/l (20 °C)	
Partitioning coefficient n-octanol/water (log Pow):	-4 (25 °C)	(OECD Guideline 107)
Adsorption/water - soil:	Study scientifically not justified.	
Viscosity, dynamic:	not applicable	
<u>Particle characteristics</u>		
Particle size distribution:	257 µm Test substance: other TS fine particles -	(D50, ISO 13320-1)

10. Stability and Reactivity

Conditions to avoid:
 Avoid humidity.

Thermal decomposition: 500 °C

Substances to avoid:
 nitrites, nitrates, oxidizing agents, acids

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Hazardous reactions:

Reacts with nitrites. Reacts with nitrates. Reacts with oxidizing agents. Generation of sulphur dioxide upon exposure to acids. (or conditions.) The product consumes oxygen.

Hazardous decomposition products:

Sulphur dioxide

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:

LD50rat (oral): approx. 2,610 mg/kg (OECD Guideline 401)

Acute inhalation toxicity

LC50 rat (by inhalation): > 5.5 mg/l 4 h (OECD Guideline 403)

No mortality was observed. Tested as dust aerosol.

Acute dermal toxicity

LD50 rat (dermal): > 2,000 mg/kg (OECD Guideline 402)

No mortality was observed.

Assessment of acute toxicity

Of low toxicity after single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. Many individuals are sensitive to sulphite additives and may experience a range of symptoms, including dermatitis, urticaria, angio-oedema, abdominal pain, diarrhoea, bronchoconstriction and anaphylaxis.

Irritation

Assessment of irritating effects:

Not irritating to eyes and skin.

Experimental/calculated data:

Skin corrosion/irritation rabbit: non-irritant (Draize test)

Skin corrosion/irritation rabbit: non-irritant (similar to OECD guideline 404)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Serious eye damage/irritation rabbit: non-irritant (Draize test)

Respiratory/Skin sensitization

Assessment of sensitization:

Skin sensitizing effects were not observed in animal studies. A sensitizing effect on particularly sensitive individuals cannot be excluded.

Experimental/calculated data:

Mouse Local Lymph Node Assay (LLNA) mouse: Non-sensitizing. (OECD Guideline 429)

Germ cell mutagenicity

Assessment of mutagenicity:

The substance was not mutagenic in bacteria. The substance was not mutagenic in a test with mammals.

Carcinogenicity

Assessment of carcinogenicity:

In long-term studies in rats in which the substance was given by feed, a carcinogenic effect was not observed. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Reproductive toxicity

Assessment of reproduction toxicity:

The results of animal studies gave no indication of a fertility impairing effect. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The chemical structure does not suggest a specific alert for such an effect.

Developmental toxicity

Assessment of teratogenicity:

No indications of a developmental toxic / teratogenic effect were seen in animal studies. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Experiences in humans

Experimental/calculated data:

With sensitive persons it can lead to an over sensitive reaction.

Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Repeated oral uptake of the substance did not cause substance-related effects. Repeated inhalative uptake of the substance did not cause substance-related effects. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aspiration hazard

not applicable

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Other relevant toxicity information

Contact with acids liberates toxic gases.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms. The inhibition of the degradation activity of activated sludge is not anticipated when introduced to biological treatment plants in appropriate low concentrations.

Toxicity to fish:

LC50 (96 h) 316 mg/l, *Leuciscus idus* (DIN 38412 Part 15, static)

The product has not been tested. The statement has been derived from substances/products of a similar structure or composition. The details of the toxic effect relate to the nominal concentration.

Aquatic invertebrates:

EC50 (48 h) 59 mg/l, *Daphnia magna* (Directive 79/831/EEC, static)

The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

EC50 (48 h) 230 mg/l, *Daphnia magna* (other, static)

The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Aquatic plants:

EC50 (72 h) 31.9 mg/l (growth rate), *Scenedesmus subspicatus* (OECD Guideline 201, static)

The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

EC50 (72 h) > 100 mg/l (growth rate), *Pseudokirchneriella subcapitata* (OECD Guideline 201, static)

Nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

Microorganisms/Effect on activated sludge:

EC50 (3 h) > 1,000 mg/l, (OECD Guideline 209, static)

The details of the toxic effect relate to the nominal concentration.

EC10 (17 h) 260 mg/l, *Pseudomonas putida* (DIN 38412 Part 8, aquatic)

Nominal concentration.

Chronic toxicity to fish:

No observed effect concentration (34 d) 316 mg/l, *Danio rerio* (OECD Guideline 210, Flow through.)

The details of the toxic effect relate to the nominal concentration.

Chronic toxicity to aquatic invertebrates:

No observed effect concentration (21 d), > 10 mg/l, *Daphnia magna* (OECD Guideline 211, semistatic)

The details of the toxic effect relate to the nominal concentration. The product has not been tested. The statement has been derived from substances/products of a similar structure or composition.

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Assessment of terrestrial toxicity:
No data available.
Study scientifically not justified.

Mobility

Assessment transport between environmental compartments:
Adsorption to solid soil phase is not expected.

Persistence and degradability

Elimination information:
Study scientifically not justified.

Assessment of stability in water:
According to structural properties, hydrolysis is not expected/probable.

Information on Stability in Water (Hydrolysis):
Study scientifically not justified.

Bioaccumulation potential

Assessment bioaccumulation potential:
Significant accumulation in organisms is not to be expected.

Bioaccumulation potential:
Study scientifically not justified.

Additional information

Other ecotoxicological advice:
Higher concentrations of the substance may cause a strong chemical oxygen consumption in biological sewage-treatment plants and/or waterways. Inhibition of degradation activity in activated sludge is not to be anticipated during correct introduction of low concentrations.

13. Disposal Considerations

Must be disposed of or incinerated in accordance with local regulations.
Observe national and local legal requirements.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible; then it can be passed on for recycling after being thoroughly cleaned.

14. Transport Information

Domestic transport:

UN number or ID number	Not classified as a dangerous good under transport regulations
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable

BASF Safety data sheet
Date / Revised: 02.12.2025
Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

Special precautions for user None known

Sea transport

IMDG

Not classified as a dangerous good under transport regulations

UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

Not classified as a dangerous good under transport regulations

UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Other regulations

If other regulatory information applies that is not already provided elsewhere in this safety data sheet, then it is described in this subsection.

Registration status:

AICS, AU	released / listed
AICS, AU	released / exempt

BASF Safety data sheet

Date / Revised: 02.12.2025

Product: **Sodium Sulfite anhydrous food grade (E221)**

Version: 1.0

(30042389/SDS_GEN_AU/EN)

Date of print: 03.12.2025

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.